Chapter 7. The Bootstrap

Sampling distributions, bootstrap estimates of standard error and bias, resampling schemes

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7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat{\theta} = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat{\theta}$, and with it the bias, standard error and quantiles, depends on the unknown F. If we could draw new samples from F we would simply simulate. The bootstrap replaces F by an estimate $\hat{F}$ and simulates from $\hat{F}$:

$$\text{real world: } F \to \mathbb{X}_n \to \hat{\theta}, \qquad \text{bootstrap world: } \hat{F} \to \mathbb{X}_n^* \to \hat{\theta}^*.$$

The distribution of $\hat{\theta}^*$ given the data, $G^* = G_{\hat{F}}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat{F} = F_n$, the empirical distribution (mass $1/n$ at each observation). Sampling from F_n means drawing n observations *with replacement* from the data.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$; their empirical distribution approximates G^*.

There are two errors: $F_n \neq F$ (statistical, of order $n^{-1/2}$, the important one) and $B < \infty$ (Monte Carlo, of order $B^{-1/2}$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat{\theta})$ and $\text{bias}_F(\hat{\theta}) = E_F\hat{\theta} - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat{\theta}^{*b} - \bar{\theta}^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar{\theta}^* - \hat{\theta}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_b \hat{\theta}^{*b}.$$

Note that the role of the true $\theta(F)$ is played by $\theta(F_n)$, which is $\hat{\theta}$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat{\theta}$ = median = 4. Among the $3^3 = 27$ equally likely resamples, the median equals the smallest value 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$. Likewise it equals 9 with probability 7/27, and 4 with probability 13/27. Hence

$$E^*\hat{\theta}^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bootstrap bias estimate is 0.52. With B random resamples these values are only approximated. Here they can be computed exactly because n is tiny.

When the bootstrap works

For smooth statistics (means, smooth functions of means, many M- and U-statistics) the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ consistently estimates that of $\sqrt{n}(\hat{\theta} - \theta)$. It can fail for non-smooth targets at the boundary of the parameter space, e.g. $\hat{\theta} = \max_i X_i$ for Uniform$(0, \theta)$, where $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$, whereas $P(\hat{\theta} = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ belongs to a parametric family, estimate $\hat{\eta}$ (e.g. by maximum likelihood) and draw bootstrap samples from $F_{\hat{\eta}}$ instead of F_n. When the model is right this is more accurate than the nonparametric bootstrap, and it is useful when the sampling distribution of $\hat{\theta}$ is analytically intractable (e.g. a nonlinear functional of a Weibull or gamma fit). When the model is wrong it can be badly biased.

7.4 Resampling designs

The guiding rule is: *resample the way the data were generated.*

- **Several samples.** For independent samples from $F_1, \ldots, F_k$, resample within each sample separately, keeping the sample sizes $n_1, \ldots, n_k$.
- **Bivariate data, random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample the pairs (**case** or pairs bootstrap). This keeps the dependence between X and Y and is valid even if a regression model is wrong.
- **Regression, fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, resample residuals: $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, with r_i^* drawn from the (leverage-adjusted) residuals. This is more efficient when the model is right but relies on it, including homoscedasticity.

Choosing between the two regression schemes:

- Classical regression inference is conditional on the x's. The residual bootstrap keeps the x's fixed, so it reproduces this setting and agrees closely with normal theory when that theory is adequate.
- The case bootstrap remains valid when the linear model is wrong. Then β is reinterpreted as the best linear approximation, the minimizer of $E(Y - \alpha - \beta X)^2$, and heteroscedasticity is allowed. The cost is some efficiency when the model is right.
- In designed experiments the x's are chosen by the experimenter, which argues for keeping them fixed.

Contrast with permutation tests: the bootstrap samples *with* replacement and keeps the structure of the alternative (e.g. the pairs (X_i, Y_i) stay together). A permutation test samples *without* replacement and deliberately destroys the association, to produce the null distribution.